

PIARCO INTL, Trinidad and Tobago

WMO: 789700

Lat: 10.595N

Lon: 61.337W

Elev: 40

StdP: 14.67

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 11634

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	69.6	71.2	64.5	91.1	84.0	66.0	96.2	82.6	21.0	85.2	19.0	85.1	0.2	90	0.359

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	13.9	93.2	78.4	91.8	77.9	91.4	77.8	80.9	88.1	80.3	87.6	79.9	87.1	11.4	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.1	151.1	84.5	78.7	149.0	84.1	77.8	144.5	83.0	44.6	88.5	44.0	88.2	43.5	87.7	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.0	17.8	16.3	67.1	95.8	1.4	1.8	66.1	97.0	65.3	98.0	64.5	99.0	63.5	100.3		
(5)			65.8	83.3	1.3	1.4	64.8	84.4	64.0	85.2	63.3	86.0	62.3	87.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.8	79.7	79.9	81.0	82.4	83.0	82.1	82.2	82.7	83.2	82.7	81.6	80.5
	DBStd	2.01	1.33	1.69	1.79	1.81	1.47	1.66	1.59	1.63	1.63	1.87	1.69	1.54
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11590	920	837	962	972	1024	964	998	1014	995	1013	947	944
	CDD65	6115	455	417	497	522	559	514	533	549	545	548	497	479
	CDH74	58610	3726	3492	4635	5336	5941	5116	5290	5513	5606	5396	4499	4060
	CDH80	21577	1277	1240	1815	2149	2291	1709	1883	2115	2215	2016	1533	1334
Wind	WSAvg	6.1	6.4	7.5	7.7	7.9	8.1	7.4	5.7	4.3	4.1	4.3	4.7	5.6
Precipitation	PrecAvg	69.5	3.1	1.7	1.5	1.9	4.0	9.8	10.1	9.5	7.3	7.4	8.9	5.9
	PrecMax	86.3	9.7	6.5	5.3	8.5	10.2	18.8	15.2	17.7	12.9	19.2	17.4	13.0
	PrecMin	57.3	0.4	0.1	0.0	0.2	0.5	3.3	4.6	4.3	2.2	1.4	2.6	1.9
	PrecStd	8.0	2.2	1.2	1.2	1.7	2.4	3.3	2.6	3.1	2.2	3.4	3.4	2.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.8	90.9	92.9	93.5	93.2	91.7	91.8	93.5	94.3	93.6	91.7	90.0
		MCWB	75.2	75.5	76.0	77.0	76.5	77.6	78.3	79.2	79.2	79.4	78.7	77.3
	2%	DB	88.1	89.3	91.2	91.7	91.6	90.0	91.1	91.8	92.9	91.8	90.3	89.3
		MCWB	75.2	75.1	75.3	76.3	76.6	77.5	78.2	78.9	79.1	79.1	78.5	77.1
	5%	DB	87.6	87.9	89.6	90.8	90.9	89.4	89.7	91.0	91.4	91.1	89.4	87.9
		MCWB	75.2	74.7	75.2	76.0	76.7	77.5	77.9	78.7	78.9	79.0	78.4	76.9
	10%	DB	86.2	86.4	88.0	89.5	89.5	87.9	88.2	89.5	89.8	89.5	87.8	86.4
		MCWB	75.1	74.4	74.9	75.8	76.8	77.7	78.1	78.5	78.8	78.8	78.6	76.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.9	78.4	78.6	79.8	80.6	80.6	80.9	81.4	81.6	81.4	81.0	80.0
		MCDB	85.0	86.0	85.7	87.2	86.7	86.7	87.2	89.1	89.0	89.1	87.6	86.3
	2%	WB	77.8	77.1	77.5	78.8	79.7	79.8	80.0	80.4	80.6	80.6	80.1	79.2
		MCDB	84.4	84.8	85.4	85.9	85.8	85.9	86.7	88.2	88.6	88.4	86.8	85.5
	5%	WB	77.0	76.3	76.7	78.0	79.0	79.2	79.4	79.9	80.0	80.0	79.6	78.4
		MCDB	84.0	83.8	84.9	85.3	85.1	85.2	86.2	87.5	87.8	87.5	86.2	84.6
	10%	WB	76.2	75.5	76.1	77.3	78.3	78.7	78.8	79.3	79.5	79.4	78.9	77.7
		MCDB	82.9	83.0	84.1	84.8	84.4	84.5	85.6	86.7	87.1	86.6	85.5	83.9
Mean Daily Temperature Range	5% DB	MDBR	14.8	15.1	15.8	15.2	13.9	12.3	13.2	14.5	15.0	14.4	13.6	13.8
		MCDBR	16.1	16.4	17.2	16.6	15.4	13.9	14.9	16.3	16.6	16.5	15.3	15.3
		MCWBR	5.7	5.3	5.0	4.3	4.2	4.5	5.3	5.8	5.8	6.0	5.8	5.7
	5% WB	MCDBR	13.9	14.6	15.1	14.3	12.9	12.0	13.3	14.8	15.2	14.8	13.9	13.4
	MCWBR	5.8	5.4	5.1	4.5	4.6	4.8	5.5	6.0	5.9	5.9	5.9	5.8	
Clear-Sky Solar Irradiance	taub	0.385	0.407	0.405	0.454	0.471	0.515	0.489	0.436	0.420	0.395	0.385	0.398	
	taud	2.533	2.439	2.456	2.316	2.270	2.123	2.219	2.417	2.451	2.550	2.575	2.520	
	Ebn at Noon	293	289	290	273	263	248	256	274	281	289	291	286	
	Edh at Noon	33	37	37	43	43	49	45	38	37	33	31	33	
All-Sky Solar Radiation	RadAvg	1617	1793	1875	1885	1792	1669	1721	1781	1792	1644	1498	1467	
	RadStd	89	79	104	85	128	80	88	75	79	69	99	89	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.45	N/A	N/A	+0.68	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	+0.36	+0.59	+0.74	+0.25	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page